

Experiment 20

Title

Write a Prolog Program for Planets Database

Aim

To implement a Planets database using Prolog facts and retrieve the properties of a planet using queries.

Objective

To understand how Prolog stores planetary information and displays the required details based on the given planet name.

Algorithm

Step 1: Define facts for each planet with its type, size, temperature, and position.

Step 2: Create a rule to retrieve the details of a selected planet.

Step 3: Accept the planet name through a query.

Step 4: Search the database for the matching planet.

Step 5: Display the planet's properties such as type, size, temperature, and position.

Program

```
% Planet Database
```

```
planet(mercury, rocky, small, hot, first).
```

```
planet(venus, rocky, small, hot, second).
```

```
planet(earth, rocky, medium, moderate, third).
```

```
planet(mars, rocky, small, cold, fourth).
```

```
planet(jupiter, gas_giant, large, cold, fifth).
```

```
planet(saturn, gas_giant, large, cold, sixth).
```

```
planet(uranus, ice_giant, large, cold, seventh).
```

```
planet(neptune, ice_giant, large, cold, eighth).
```

```
% Display Planet Information
```

show_planet(Name) :-

```
planet(Name, Type, Size, Temperature, Position),  
write('Planet Name : '), write(Name), nl,  
write('Type      : '), write(Type), nl,  
write('Size      : '), write(Size), nl,  
write('Temperature : '), write(Temperature), nl,  
write('Position   : '), write(Position), nl.
```

Sample Input

?- show_planet(earth).

Sample Output

```
| ?- consult('C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp20.pl').  
compiling C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp20.pl for byte code...  
C:/Users/B SANTHOSH KUMAR/OneDrive/Documents/exp20.pl compiled, 20 lines read - 2866 bytes written, 7 ms  
  
(16 ms) yes  
| ?- show_planet(earth).  
Planet Name : earth  
Type        : rocky  
Size        : medium  
Temperature : moderate  
Position    : third
```

Result

Hence, the Prolog program for the Planets Database was executed successfully and the correct output was obtained.